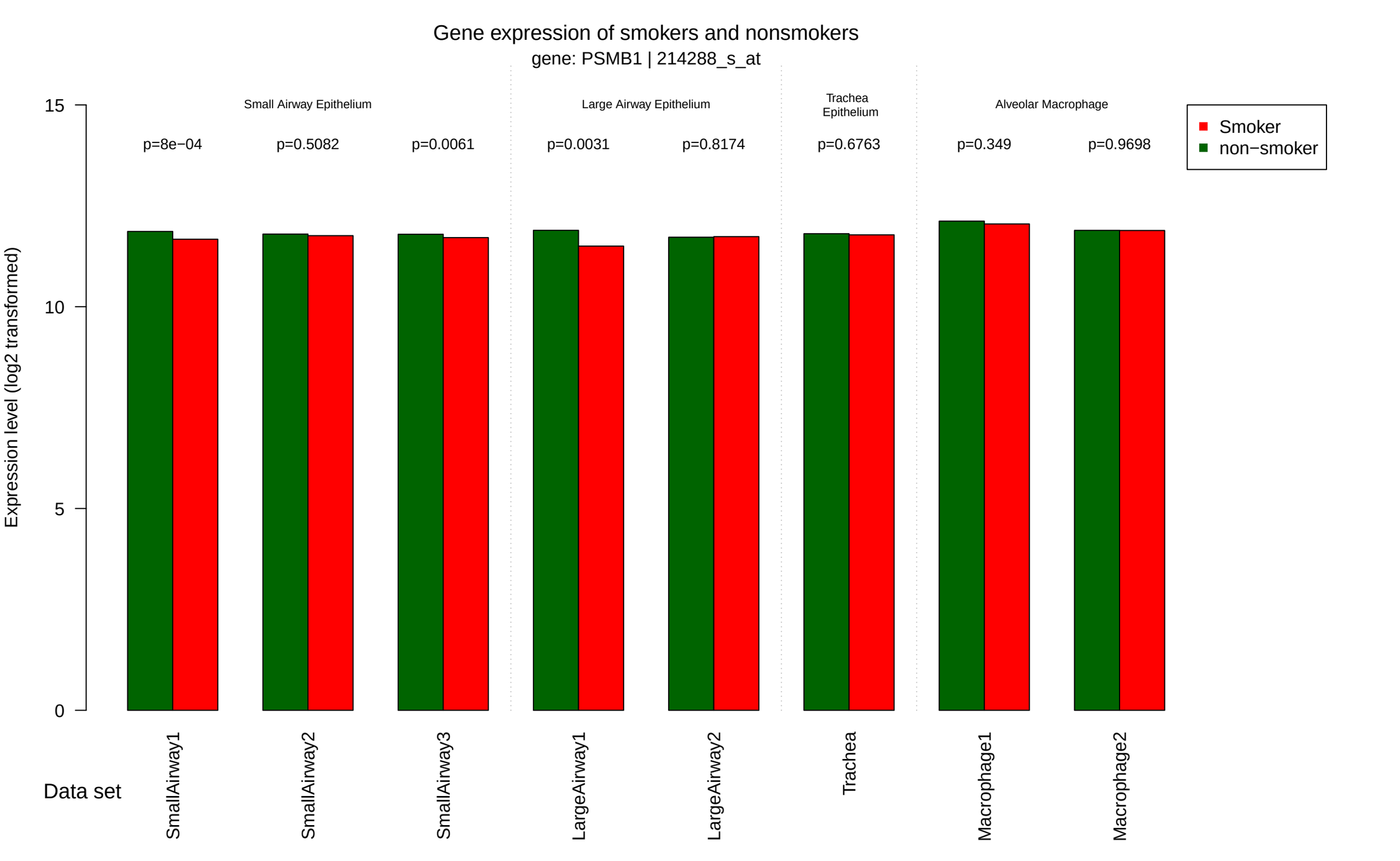
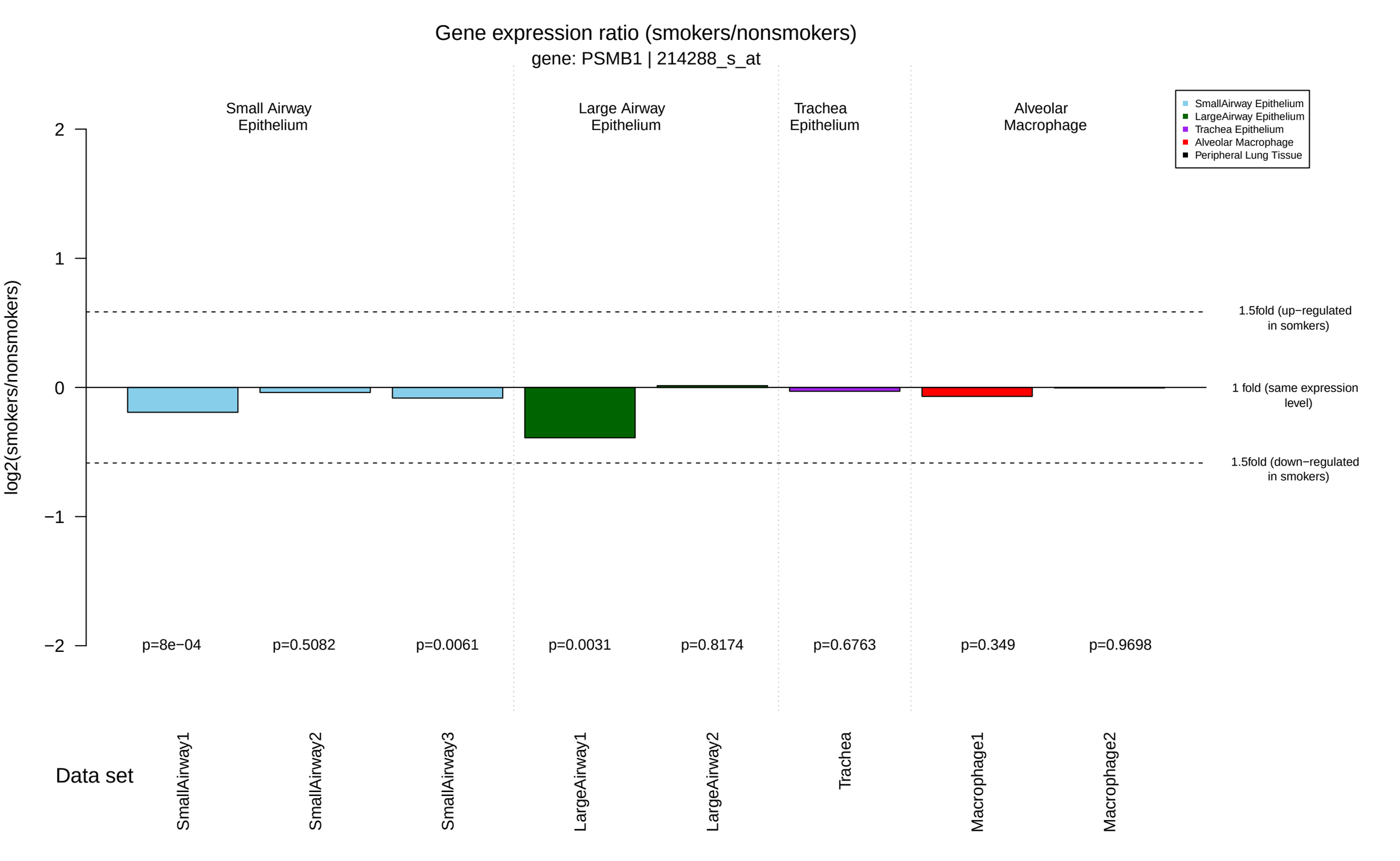
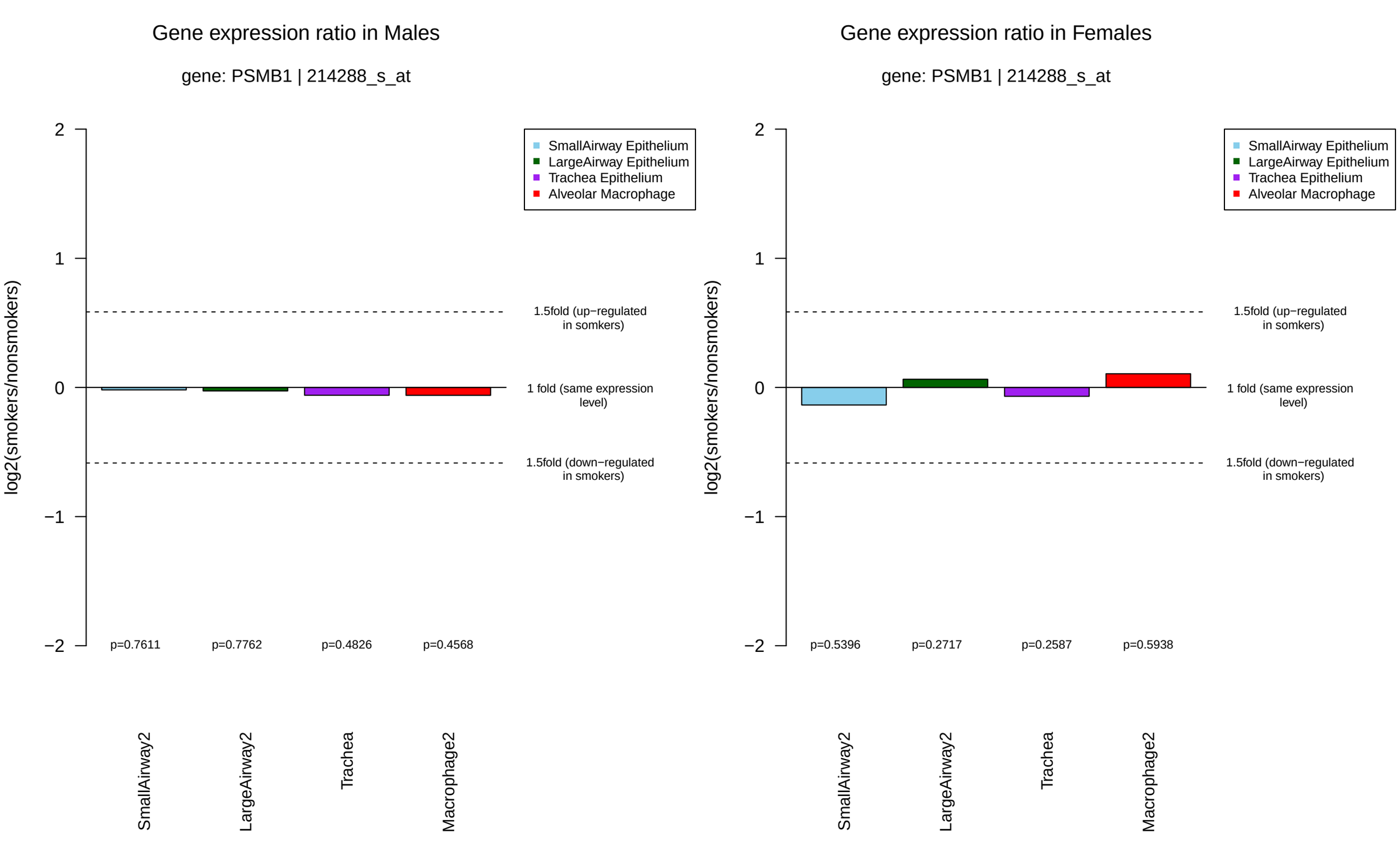
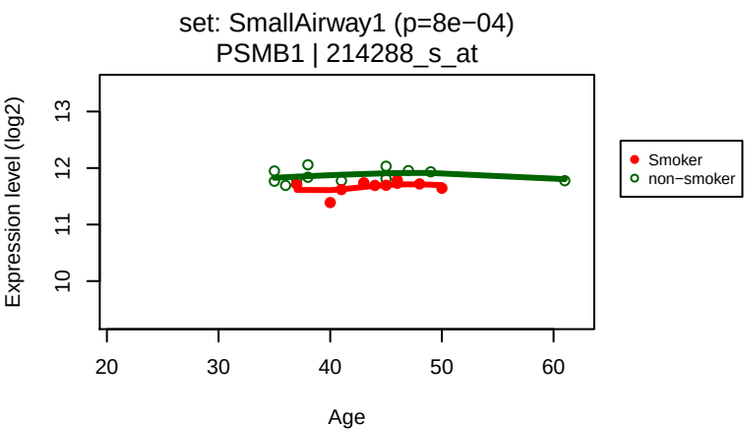
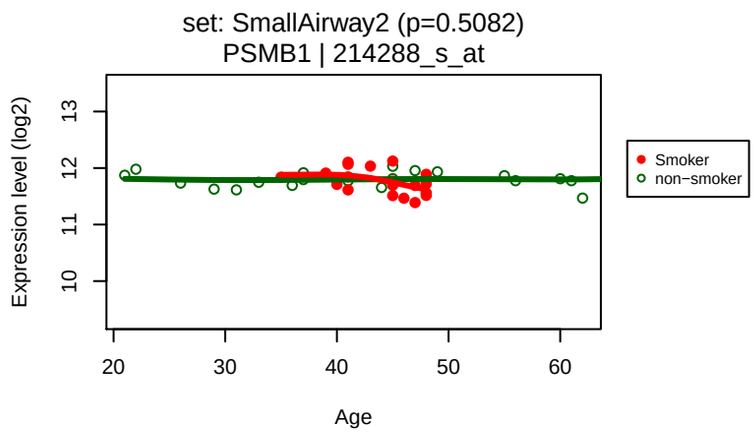
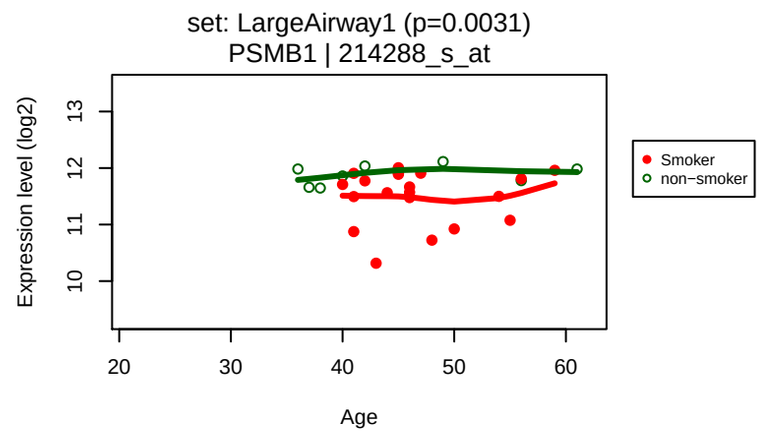

Gene expression of smokers and nonsmokers
gene: PSMB1 | 214288_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=8e−04
p=0.5082
p=0.0061
p=0.0031
p=0.8174
p=0.6763
p=0.349
p=0.9698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

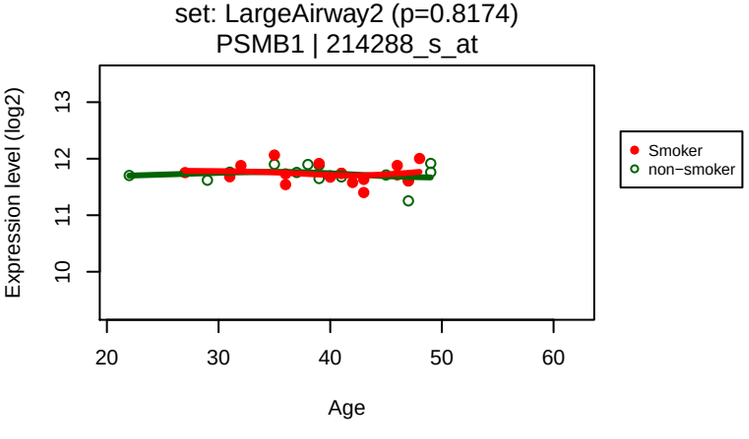

Gene expression ratio (smokers/nonsmokers)
gene: PSMB1 | 214288_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=8e−04
p=0.5082
p=0.0061
p=0.0031
p=0.8174
p=0.6763
p=0.349
p=0.9698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

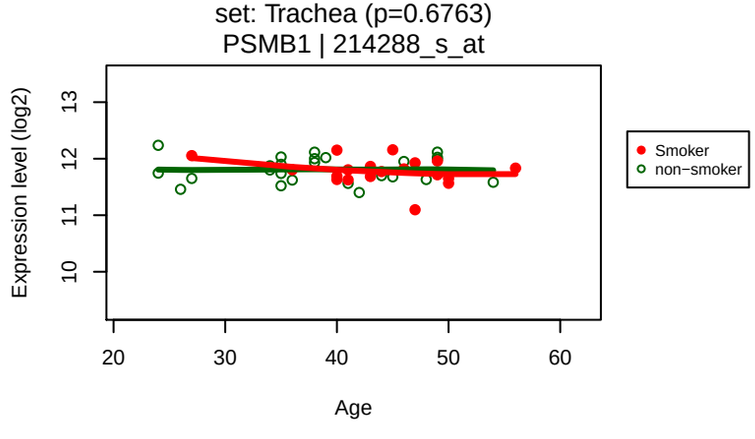

Gene expression ratio in Males
gene: PSMB1 | 214288_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7611
p=0.7762
p=0.4826
p=0.4568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB1 | 214288_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5396
p=0.2717
p=0.2587
p=0.5938
SmallAirway2
LargeAirway2
Trachea
Macrophage2

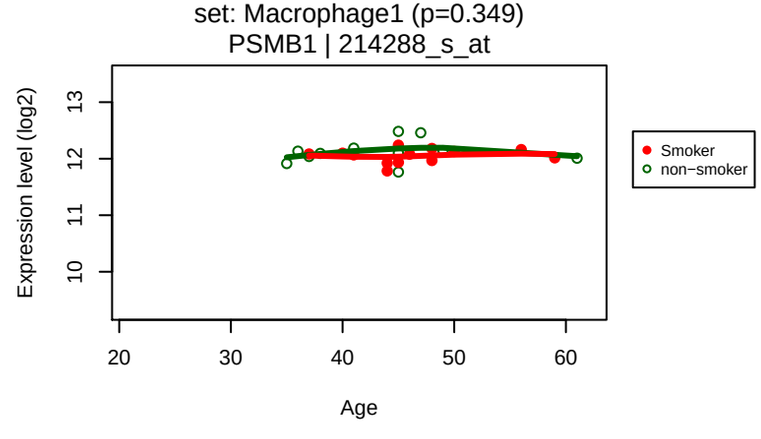

set: SmallAirway1 (p=8e−04)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker

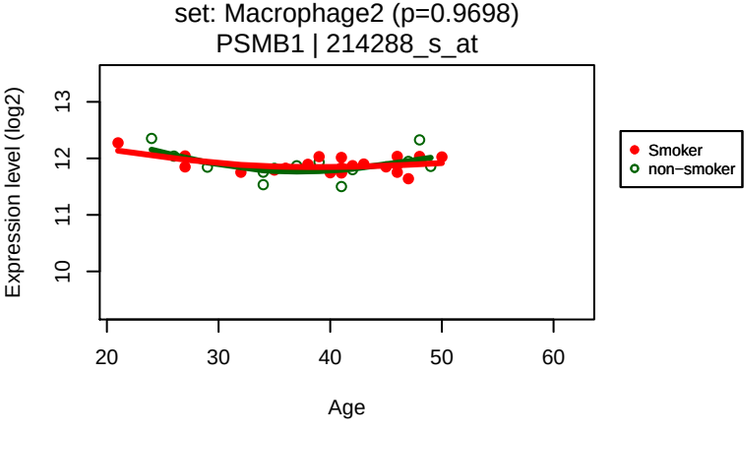

set: SmallAirway2 (p=0.5082)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker

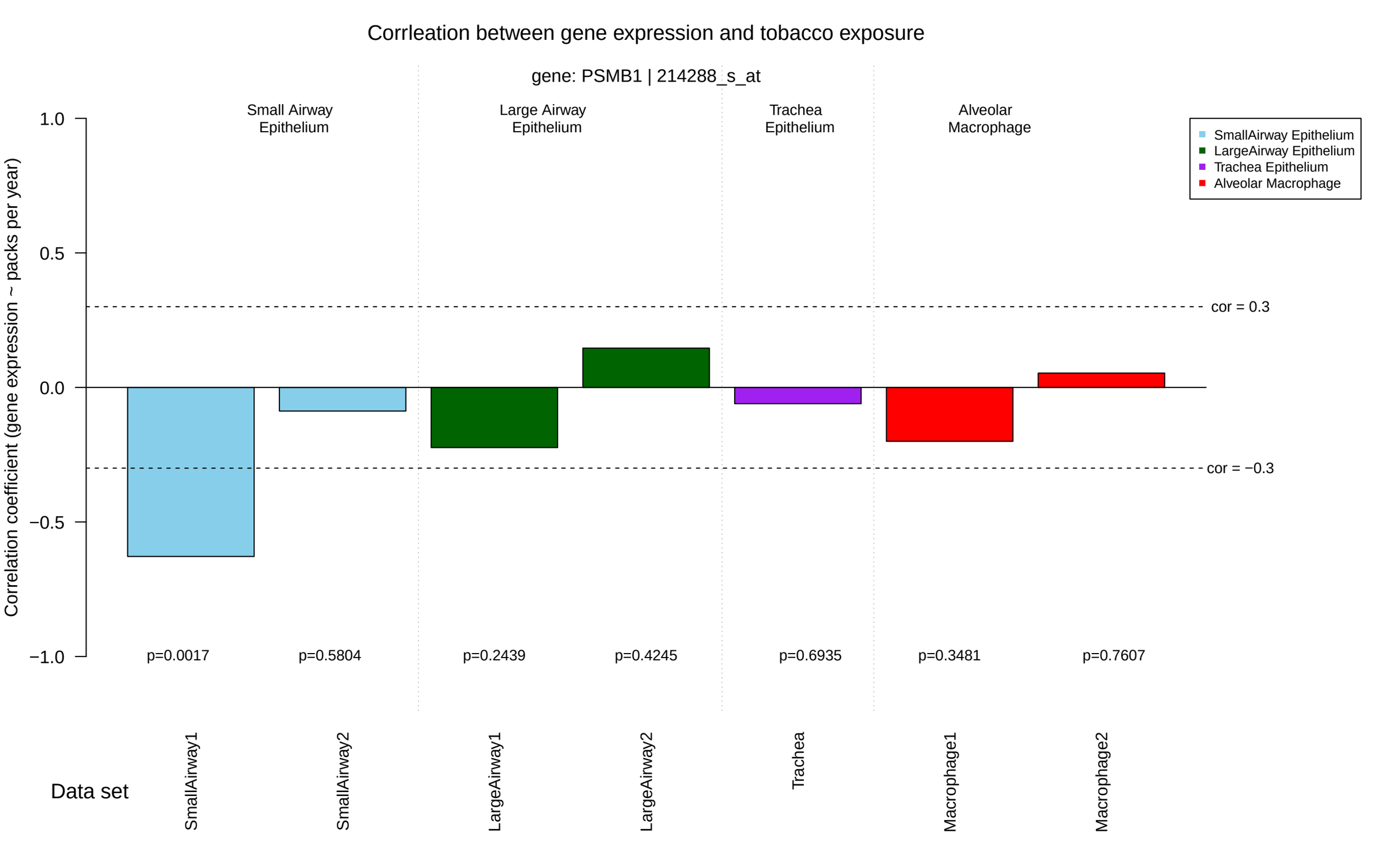

set: LargeAirway1 (p=0.0031)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8174)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6763)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.349)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9698)
PSMB1 | 214288_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB1 | 214288_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0017
p=0.5804
p=0.2439
p=0.4245
p=0.6935
p=0.3481
p=0.7607
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2